– Gram-Schmidt Yöntemi Soru Çözüm –

1) $(\mathbb{R}^3, \langle .,. \rangle_E)$ iç çarpım uzayı olmak üzere $\{\underbrace{(1,2,1)}_{u_1}, \underbrace{(2,1,0)}_{u_2}, \underbrace{(3,3,2)}_{u_3}\}$ tabanını Gram-Schmidt yöntemi ile ortonormalleştiriniz.

Çözüm :

$\{v_1, v_2, v_3\}$ ortogonal taban olsun.

1. adım : $v_1 = (1,2,1)$

2 adım :

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = (2,1,0) - \frac{\langle (2,1,0),(1,2,1) \rangle}{\|(1,2,1)\|^2} \cdot (1,2,1)$$

$$= (2,1,0) - \frac{\cancel{4}^{2}}{\cancel{6}_{3}}(1,2,1) = (2,1,0) - \left(\frac{2}{3}, \frac{4}{3}, \frac{2}{3}\right)$$

$$= \left(\frac{4}{3}, \frac{-1}{3}, \frac{-2}{3}\right)$$

3 adım :

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$\Rightarrow v_3 = (3,3,2) - \frac{\langle (3,3,2),(1,2,1) \rangle}{\|(1,2,1)\|^2} \cdot (1,2,1) - \frac{\langle (3,3,2), \left(\frac{4}{3}, \frac{-1}{3}, \frac{-2}{3}\right) \rangle}{\left\|\left(\frac{4}{3}, \frac{-1}{3}, \frac{-2}{3}\right)\right\|^2} \cdot \left(\frac{4}{3}, \frac{-1}{3}, \frac{-2}{3}\right)$$

$$= (3,3,2) - \frac{11}{6} \cdot (1,2,1) - \frac{5/\cancel{3}}{21/\cancel{9}_{3}} \cdot \left(\frac{4}{3}, \frac{-1}{3}, \frac{-2}{3}\right)$$

$$= (3,3,2) - \left(\frac{11}{6}, \frac{22}{6}, \frac{11}{6}\right) - \left(\frac{20}{21}, \frac{-5}{21}, \frac{-10}{21}\right) = \left(\frac{3}{14}, \frac{-3}{7}, \frac{9}{14}\right)$$

$$\{v_1, v_2, v_3\} = \left\{(1,2,1), \left(\frac{4}{3}, \frac{-1}{3}, \frac{-2}{3}\right), \left(\frac{3}{14}, \frac{-3}{7}, \frac{9}{14}\right)\right\} \rightarrow \text{ortogonal!}$$

$$\Rightarrow \left\{ \left(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right), \left(\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}\right), \left(\frac{3}{\sqrt{126}}, \frac{-6}{\sqrt{126}}, \frac{9}{\sqrt{126}}\right) \right\}$$

ortonormal taban!

2) $\langle u,v\rangle = u_1.v_1 + 2u_2v_2 + u_3.v_3$ iç çarpımı ile donatılmış $(\mathbb{R}^3, \langle .,. \rangle)$ için $\{\underbrace{(1,0,1)}_{u_1}, \underbrace{(1,1,0)}_{u_2}, \underbrace{(0,2,0)}_{u_3}\}$ tabanını Gram-Schmidt yöntemi ile ortonormalleştiriniz.

Çözüm :

$\{v_1,v_2,v_3\}$ ortogonal bir taban olsun

$\|u\| = \sqrt{\langle u,u\rangle}$

$\boxed{\|u\|^2 = \langle u,u\rangle}$

$1.1 + 2.1.0 + 0.1 = 1$

- $v_1 = u_1 = (1,0,1)$

- $v_2 = u_2 - \dfrac{\langle u_2,v_1\rangle}{\|v_1\|^2}.v_1 = (1,1,0) - \dfrac{\langle (1,1,0),(1,0,1)\rangle}{\|(1,0,1)\|^2}.(1,0,1)$

$\langle (1,0,1),(1,0,1)\rangle = 2$

$= (1,1,0) - \dfrac{1}{2}(1,0,1)$

$= \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$

- $v_3 = u_3 - \dfrac{\langle u_3,v_1\rangle}{\|v_1\|^2}v_1 - \dfrac{\langle u_3,v_2\rangle}{\|v_2\|^2}.v_2$

$0 + 2.2.1 + 0$

$= (0,2,0) - \dfrac{\langle (0,2,0),(1,0,1)\rangle}{\|(1,0,1)\|^2}.(1,0,1) - \dfrac{\langle (0,2,0),(1/2,1,-1/2)\rangle}{\|(1/2,1,-1/2)\|^2}.(1/2,1,-1/2)$

$\langle (\frac{1}{2},1,-\frac{1}{2}),(\frac{1}{2},1,-\frac{1}{2})\rangle$

$\frac{1}{4} + 2.1 + \frac{1}{4}$

$\frac{1}{2} + 2$

$= (0,2,0) - \cancel{\dfrac{0}{2}(1,0,1)} - \dfrac{4}{\frac{5}{2}}\left(\frac{1}{2},1,-\frac{1}{2}\right)$

$= (0,2,0) - \frac{8}{5}.\left(\frac{1}{2},1,-\frac{1}{2}\right)$

$= (0,2,0) - \left(\frac{4}{5}, \frac{8}{5}, -\frac{4}{5}\right) = \left(-\frac{4}{5}, \frac{2}{5}, \frac{4}{5}\right)$

$\left\{(1,0,1), \left(\frac{1}{2}, 1, -\frac{1}{2}\right), \left(-\frac{4}{5}, \frac{2}{5}, \frac{4}{5}\right)\right\} \longrightarrow$ ortogonal!

$\|(1,0,1)\| = \sqrt{\langle (1,0,1)(1,0,1)\rangle} = \sqrt{2}$

$\|(\frac{1}{2},1,-\frac{1}{2})\| = \sqrt{\frac{5}{2}}$

$\|(-\frac{4}{5},\frac{2}{5},\frac{4}{5})\| = \sqrt{\langle(-\frac{4}{5},\frac{2}{5},\frac{4}{5}).(-\frac{4}{5},\frac{2}{5},\frac{4}{5})\rangle} = \sqrt{\frac{40}{25}} = \frac{2\sqrt{10}}{5}$

$\left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{\sqrt{2}}{2\sqrt{5}}, \frac{\sqrt{2}}{\sqrt{5}}, \frac{-\sqrt{2}}{2\sqrt{5}}\right), \left(\frac{-2}{\sqrt{10}}, \frac{1}{\sqrt{10}}, \frac{2}{\sqrt{10}}\right)\right\}$ → ortonormal!

$$\left\| \left(-\tfrac{4}{5}, \tfrac{2}{5}, \tfrac{4}{5}\right) \right\| = \sqrt{\left\langle \left(-\tfrac{4}{5}, \tfrac{2}{5}, \tfrac{4}{5}\right), \left(-\tfrac{4}{5}, \tfrac{2}{5}, \tfrac{4}{5}\right) \right\rangle} = \sqrt{\frac{40}{25}} = \frac{2\sqrt{10}}{5}$$

3) P_2 uzayı için $\langle p,q\rangle = \int_{-1}^{1} p(x)\cdot q(x)\cdot dx$ iç çarpımı verilsin. Buna göre $\{\underset{u_1}{1}, \underset{u_2}{x}, \underset{u_3}{x^2}\}$ tabanına karşılık gelen ortonormal tabanı bulunuz.

$\{p_1, p_2, p_3\}$ ortogonal taban olsun

$$\|u\| = \sqrt{\langle u,u\rangle}$$

$$\boxed{\|u\|^2 = \langle u,u\rangle}$$

- $p_1 = u_1 = 1$

- $$p_2 = u_2 - \frac{\langle u_2, p_1\rangle}{\|p_1\|^2} p_1 = x - \frac{\int_{-1}^{1} x\,1\,dx}{\int_{-1}^{1} 1\cdot 1\cdot dx}\cdot 1$$

$$= x - \frac{0}{2}\cdot 1 = x$$

- $$p_3 = u_3 - \frac{\langle u_3, p_1\rangle}{\|p_1\|^2} p_1 - \frac{\langle u_3, p_2\rangle}{\|p_2\|^2} p_2$$

$$= x^2 - \frac{\int_{-1}^{1} x^2\cdot 1\,dx}{2}\cdot 1 - \frac{\cancel{\int_{-1}^{1} x^2\cdot x\,dx}^{\,0}}{\int_{-1}^{1} x\,x\,dx}\,x$$

$$= x^2 - \frac{\frac{2}{3}}{2}\cdot 1 - 0 = x^2 - \frac{1}{3}$$

$\{p_1, p_2, p_3\} = \{1, x, x^2 - \frac{1}{3}\} \to$ ortogonal taban!

$\Rightarrow \left\{ \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}}x, \frac{\sqrt{45}}{\sqrt{28}}\cdot\left(x^2 - \frac{1}{3}\right) \right\} \to$ ortonormal!

- $\|1\| = \sqrt{\int_{-1}^{1} 1\cdot 1\cdot dx} = \sqrt{2}$

- $\|x\| = \sqrt{\int_{-1}^{1} x\cdot x\cdot dx} = \sqrt{\frac{2}{3}}$

- $\|x^2 - \frac{1}{3}\| = \sqrt{\int_{-1}^{1} \left(x^2 - \frac{1}{3}\right)^2 dx} = \sqrt{\frac{28}{45}}$